

Probe–Target Alignment in In-Context Membership Inference

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Abstract

Membership inference attacks test whether a target sample was used during model training. Recent in-context probing attacks estimate membership by measuring how a fine-tuned language model’s likelihood changes when a probe context is prepended to the target input. This project studies a focused question: does the relationship between the probe and the target sample affect the membership signal? I implement a scaled-down ICP-MIA pipeline on HealthcareMagic using Pythia-410M with LoRA fine-tuning and compare three probe constructions: self-perturbation, unrelated external reference data, and in-domain held-out reference data. The strongest result is self-perturbation (AUC 0.5193), while reference-based probing with Dolly and HealthcareMagic validation remains near random. The results do not reproduce large-model performance, but they show that probe construction, model capacity, and reference-source alignment are important practical factors for in-context membership inference.

1 Introduction

Large language models are often adapted to sensitive domains through fine-tuning. In medical, legal, educational, and enterprise settings, fine-tuning data can contain private or proprietary examples. Membership inference attacks (MIAs) study whether an adversary can determine if a particular sample was used in model training or fine-tuning. A successful MIA can expose privacy risks even when the adversary does not directly recover training data.

This project focuses on in-context probing for membership inference. In-context probing attacks use additional context at inference time and observe how the target model’s likelihood changes. The intuition is that a training member has already been optimized by fine-tuning, so helpful context may produce a smaller likelihood improvement than it does for a non-member. Prior work on ICP-MIA shows strong attack performance in large-model settings, especially when the probe context is semantically related to the target sample [1].

The central question in this project is not only whether ICP-MIA can be reproduced, but whether the probe–target relationship affects the attack signal. My original ideation proposed studying probe contexts ranging from highly aligned to unrelated. In the completed implementation, I operationalize this idea through three probe sources: self-derived masked probes, unrelated external reference probes from Dolly, and in-domain held-out medical reference probes from HealthcareMagic validation.

The contributions of this project are:

- I implement an end-to-end, scaled-down ICP-MIA pipeline including HealthcareMagic data preparation, target-model fine-tuning, perturbation generation, and attack evaluation.
- I compare self-perturbation and reference-based probes as initial alignment conditions.

- I identify and fix a perturbation-generation issue that caused empty probe lists and constant ICP scores.
- I analyze why the resulting scores remain near random and discuss how target model capacity and probe quality interact with alignment.

2 Background and Related Work

2.1 Membership Inference Attacks

Membership inference attacks determine whether a particular example was part of a model’s training data. Early work showed that overfitted classifiers can leak membership through confidence scores [2]. For language models, related attacks often use loss, likelihood, compression, or neighborhood comparison signals to distinguish members from non-members [3–5].

Loss-based attacks assume that a model tends to assign lower loss to examples it has seen during training. However, language models can assign low loss to many non-member examples that are simply easy or common. This motivates stronger methods such as Min-K%, ReCaLL, and reference-based comparisons [6]. These attacks attempt to isolate membership-specific behavior from general sample difficulty.

2.2 In-Context Probing

ICP-MIA estimates membership by comparing the likelihood of a target output with and without a probe context [1]. For a target sample $s = (x, y)$, a target model M , and a probe context C , the score can be written as:

$$\text{ICP}(s, C) = LL_M(y \mid x) - LL_M(y \mid C \oplus x). \quad (1)$$

Here, $C \oplus x$ means that the probe context is prepended before the target input. If the probe improves likelihood differently for members and non-members, the resulting score can be used for membership inference.

There are two main probe construction strategies. In ICP-MIA-SP, the probe is produced from the target sample itself through self-perturbation. In ICP-MIA-Ref, the probe comes from an external or auxiliary reference pool. The reference-based version makes probe–target alignment especially important because the selected reference example may be related or unrelated to the target.

2.3 Probe–Target Alignment

My ideation proposed that semantic alignment between the probe and target may be a critical factor. Highly aligned probes may provide useful context and reveal an optimization gap. Weakly aligned or unrelated probes may be ignored by the model, add noise, or produce a different robustness-based signal. This project studies that idea through a practical initial comparison of probe sources rather than a full continuous similarity-binning analysis.

3 Method

3.1 Threat Model

The attacker is given target samples and can evaluate the target model’s likelihood under different contexts. The attacker does not know the training set directly during real deployment, but in the experiment the researcher knows which examples are members and non-members so that AUC and low-FPR TPR can be computed. Members are samples from the fine-tuning split, and non-members are held-out samples from the test split.

3.2 Target Model Training

I used HealthcareMagic as the target domain. The dataset was split into training, validation, test, and attack files. The target model was trained only on the HealthcareMagic training split. The attack set combined known members from the training split and known non-members from the test split.

Due to local hardware limits, I did not use the original paper’s large target models. Instead, I fine-tuned EleutherAI/pythia-410m-deduped using LoRA in LLaMA-Factory. I first trained on 2,000 samples to validate the pipeline, then trained on 10,000 samples for the final reported experiments. The final target model used Pythia-410M, LoRA fine-tuning, 10,000 HealthcareMagic examples, two epochs, and a maximum prompt length of 1024 tokens.

The training process also required several engineering choices that affected the final experiment. Because the local machine used Apple MPS rather than an NVIDIA CUDA GPU, mixed precision was disabled and data-loader multiprocessing was set to zero. The LoRA adapter was then exported and merged with the base Pythia model so that the attack code could load the target model as a single Hugging Face model directory. This merge step was necessary because the attack configuration expected one target-model path, not separate base-model and adapter paths.

The resulting pipeline was therefore not a direct reproduction of the original large-model setup. Instead, it was a controlled local reproduction whose purpose was to preserve the experimental logic while keeping the computation feasible. This distinction is important: the experiment can test whether the pipeline and alignment idea are observable in a small setting, but it cannot be used to claim the same level of vulnerability reported for larger instruction-tuned models.

3.3 Probe Conditions

Table 1 summarizes the probe conditions. These conditions are designed to approximate the alignment categories in the ideation while keeping the experiment feasible.

For self-perturbation, I generated 20 masked perturbations per attack sample using a 0.5 mask rate. A key implementation issue occurred during this step. The first perturbation command used the wrong mode and created examples with the field `mask_perturbations` but with an empty list. As a result, every ICP score was exactly zero and AUC was exactly 0.5. The corrected command used the conversion mode, which produced 20 non-empty perturbations per sample.

Table 1: Probe conditions used in the alignment study.

Condition	Source	Interpretation
ICP-MIA-SP	Self-perturbation	Target-derived, highly aligned probe
ICP-MIA-Ref	Dolly	External, general-domain, weakly related probe
ICP-MIA-Ref	HealthcareMagic validation	In-domain held-out medical probe

Table 2: Attack performance on the Pythia-410M 10k HealthcareMagic target model.

Method / probe	AUC	1%	5%	10%
ICP-MIA-SP / self	0.5193	0.0000	0.0590	0.1030
ICP-MIA-Ref / Dolly	0.4877	0.0080	0.0540	0.1060
ICP-MIA-Ref / HCM-Val	0.4948	0.0160	0.0500	0.1160

For reference-based probing, I ran two reference conditions. The first used Dolly, a general external instruction dataset. The second used HealthcareMagic validation as an in-domain held-out reference pool. The validation set is not a true external dataset, so I treat it as an ablation rather than the final external-reference condition.

For reference-based probing, the attack embeds candidate reference examples and selects prefixes according to similarity to the target sample. In the Dolly condition, the retrieved prefix is external but usually weakly related to the medical target. In the HealthcareMagic validation condition, the prefix is medically in-domain but still may not be closely matched to the target patient’s symptoms or answer. This distinction matters because the ideation concerns probe–target alignment at the example level, not only at the dataset level. A dataset can be in the same broad domain while individual retrieved probes remain weak.

3.4 Evaluation Metrics

I evaluate attack performance with ROC-AUC and TPR at low FPR thresholds. AUC measures overall separability between member and non-member scores. TPR@1%FPR, TPR@5%FPR, and TPR@10%FPR measure whether the attack can identify members while keeping false positives low. I also inspect score distributions to check whether the attack is producing meaningful non-constant scores.

4 Results

4.1 Main Results

Table 2 reports the final attack results. The best AUC is obtained by self-perturbation, but all conditions remain close to random. This means the scaled-down model and local setting produce only a weak membership signal.

The self-perturbation result is the only condition above 0.5 AUC. This suggests that target-derived probes were more useful than the

Table 3: Diagnostic summary before correcting perturbation generation.

Score statistic	Value
Mean	0.0000
Standard deviation	0.0000
Minimum	0.0000
Maximum	0.0000

tested reference probes. However, the effect is small. The reference-based results show that neither unrelated external data nor in-domain validation data was sufficient to create a strong ICP signal in this setting.

The low-FPR results are also close to random. At 5% FPR, self-perturbation achieves TPR 0.0590, Dolly achieves 0.0540, and HealthcareMagic validation achieves 0.0500. These values are only slightly above the false-positive thresholds themselves. Therefore, the attacks do not provide a practically useful membership test in this small-model setting. The useful contribution of the experiment is not a strong attack result, but a diagnosis of what conditions did and did not produce a measurable signal.

The reference results are especially important for the ideation. If dataset-level domain alignment alone were sufficient, the HealthcareMagic validation reference pool should have clearly outperformed Dolly. It did not. This suggests that an alignment study should measure semantic similarity directly rather than treating dataset source as a complete proxy.

4.2 Debugging Empty Perturbations

An important outcome of the project was identifying a failure mode in the reproduction pipeline. Before the perturbation file was corrected, the detailed result file showed: This explained why the AUC was exactly 0.5. The issue was not only the small model; the attack had no actual probes to use. After regenerating the file with non-empty perturbations, the self-perturbation attack produced AUC 0.5193. This shows that the pipeline was functioning after the fix, even though the signal remained weak.

4.3 Interpretation of Reference-Based Results

The Dolly reference result was slightly below random. This is plausible because Dolly is a general instruction dataset and may not provide helpful medical context for HealthcareMagic targets. The HealthcareMagic validation condition also remained near random. This was surprising because it is in-domain. A possible explanation is that simply being in the same domain is not enough; the retrieved prefix must be specifically useful for the target output, and the target model must be strong enough to use the prefix.

This supports a more nuanced version of the ideation. Alignment may matter, but it is not automatically captured by the source dataset label. A medical reference example can still be unhelpful if it is not semantically close to the target question, if the prompt is truncated, or if the model does not use in-context examples effectively.

5 Discussion

5.1 Alignment Analysis

The three probe conditions should be interpreted as an initial operationalization of the alignment idea. Self-perturbation is the strongest form of local alignment because the probe is derived from the target output itself. It preserves many lexical and structural cues from the answer, even though half of the words are replaced by placeholders. If the target model can use these cues, the probe should change the likelihood of the original answer. This condition produced the highest AUC, which is consistent with the intuition that target-specific probes are more informative than unrelated prefixes.

Dolly represents the opposite side of the alignment spectrum. It is external and general-purpose, so it is useful for testing whether arbitrary instruction-style context can reveal membership. The result was below 0.5 AUC, indicating that these prefixes did not help distinguish members from non-members in the expected direction. This does not necessarily mean Dolly is harmful in every setting; rather, it suggests that the retrieved Dolly prefixes were not useful for HealthcareMagic medical answers under the current target model.

HealthcareMagic validation is more ambiguous. At the dataset level, it is aligned because it comes from the same medical QA distribution. At the sample level, however, a validation example about one medical condition may not help with a target example about another condition. This explains why in-domain validation did not clearly improve the result. The ideation therefore needs a more precise alignment measurement: similarity should be measured between each probe and target sample, not inferred only from the source dataset.

5.2 Score Direction and Near-Random AUC

Because some reference results were slightly below 0.5, I also considered whether the score direction was reversed. In a membership attack, a score can be informative even when its sign is opposite from the expected direction. In that case, using the negative score would produce an AUC above 0.5. This check is important because different implementations may define the ICP score as either a likelihood gain or a likelihood reduction.

In this project, the main conclusion did not rely on a strong reversed-score effect. The larger issue was that the reference-based scores did not create clear separation. The self-perturbation condition produced the best result, but the margin remained small. This suggests that the attack signal was weak rather than simply reversed.

5.3 Comparison to a Full Similarity-Binning Study

The original ideation proposed grouping probes into highly aligned, moderately aligned, weakly aligned, and unrelated categories using embedding similarity. The completed experiments approximate this design but do not fully implement it. A full version would first build a large candidate pool, compute embeddings for both targets and candidate probes, and then assign each probe to a similarity bin using cosine similarity. The attack would then be repeated with

probes from each bin while holding the target model, target samples, and attack hyperparameters fixed.

This would allow stronger claims than the current source-based comparison. For example, it would be possible to test whether AUC increases monotonically with similarity, whether there is an optimal middle range, or whether unrelated probes produce a robustness signal. The present project should therefore be viewed as a first-stage experiment that establishes the pipeline and identifies why a more systematic similarity-binning analysis is needed.

5.4 What Was Completed

This project completed a scaled-down version of the ideation. I implemented the full attack workflow, compared self-derived and reference-derived probes, and analyzed results under three alignment-related probe sources. This is not a complete similarity-controlled study, but it provides an initial empirical test of the probe–target alignment hypothesis.

The strongest practical finding is that implementation details can dominate the result. Before correcting perturbation generation, the attack produced all-zero scores. After fixing it, the scores became meaningful but weak. This shows why debugging score distributions is necessary before interpreting AUC.

5.5 Why the Scores Are Low

The most likely reason for the low scores is the target model setting. Pythia-410M is much smaller than the models used in the original ICP-MIA evaluation. In-context probing depends on the model’s ability to use additional context. A small base model fine-tuned with LoRA may not respond to probes strongly enough to expose a clear optimization gap.

Other differences also matter. The experiment used LoRA rather than full fine-tuning, 10,000 samples rather than the full training split, and a maximum prompt length of 1024 tokens. During attack execution, many prompts were truncated. These constraints were necessary for local feasibility but reduce comparability to large-model experiments.

5.6 Lessons for Practical Reproduction

A useful lesson from this project is that reproducing a privacy attack requires validating each intermediate artifact, not only the final AUC. In this project, the model trained successfully and the attack script completed, but the first detailed-score file revealed that all ICP scores were zero. That observation changed the diagnosis from a model-capacity explanation to a data-format explanation. After the perturbation file was regenerated correctly, the score distribution became non-constant and the attack returned a non-random AUC. This kind of intermediate validation is especially important for LLM privacy experiments because the pipeline contains several stages: dataset construction, fine-tuning, adapter merging, prompt construction, probe retrieval, likelihood computation, and metric calculation.

5.7 Relation to the Ideation

The original ideation proposed four similarity levels: highly aligned, moderately aligned, weakly aligned, and unrelated. My completed experiments approximate three categories: self-aligned, unrelated

external, and in-domain held-out. The missing component is continuous embedding-based similarity measurement. Therefore, this report should not claim that the full similarity-binning framework was completed. Instead, it presents an initial alignment-focused study and identifies the next steps needed to fully test the hypothesis.

5.8 Why the Test Split Was Not Used as Reference Data

The test split was reserved for non-member evaluation. Using it as a reference pool would have made the experiment less clean because the same examples could influence probe construction and also serve as non-member targets. For that reason, I used validation data for the in-domain reference ablation. This choice preserves a separation between evaluation examples and reference examples. However, validation data is still from HealthcareMagic, so it should be described as held-out in-domain data rather than a truly external medical dataset.

A more rigorous future version should use an independent medical reference corpus. That design would allow a cleaner comparison between unrelated external data, related external medical data, and in-domain validation data. It would also better match the original ideation’s goal of studying whether probe–target similarity, rather than dataset identity, controls the ICP signal.

6 Conclusion and Future Work

This project investigated probe–target alignment in in-context membership inference. I reproduced the ICP-MIA workflow in a scaled-down setting, trained a HealthcareMagic target model, generated self-perturbation probes, ran reference-based probing, and evaluated member/non-member separability. The main empirical result is that self-perturbation achieved the highest AUC (0.5193), while reference-based probing with Dolly and HealthcareMagic validation remained near random.

The results suggest that probe construction matters, but they do not prove that in-domain or semantically related references always improve ICP-MIA. Future work should use a stronger target model such as Pythia-2.8B or LLaMA-3B, full or larger fine-tuning data, longer prompt windows, and a true external medical reference dataset such as iCliniq or MedInstruct. The most important extension is to compute embedding similarity between probes and targets, group probes into similarity bins, and measure whether AUC changes systematically across those bins.

Acknowledgments

This project was completed as part of a graduate privacy course. The implementation built on the public ICP-MIA repository and LLaMA-Factory for target-model fine-tuning.

References

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A Selected Commands

The corrected perturbation step used the `convert` mode instead of the `single` mode. The important command structure was:

```
python generate_perturbations.py convert \  
--input <attack.json> \  
--output <attack_perturbed.json> \  
--mask_rate 0.5 \  
--num_perturbations 20
```

The attack was then executed with the Healthcare ICP-MIA configuration:

```
python icp_mia_attack.py --config <config.yaml>
```